



Elementary Concepts of Electrical & Electronics Engineering

This course links electrical energy consumption and conservation with practical energy-conversion applications, magnetic circuits and bipolar-junction-transistor operation and biasing. Numerical methods are emphasised alongside device principles.

REV2026

Semester II

Theory

Malayalam support

Printable handbook

Course code	2031
Semester	II
Course type	Theory
Contact hours	60
Credits	4
CIE / ESE	40 / 60
Modules	4

COURSE OVERVIEW

What this handbook covers

The structure follows the supplied official SITTTTR Kerala Revision 2026 syllabus. It is a learning handbook rather than a replacement for the official syllabus PDF. Use the module checklist to ensure that every prescribed point has been revised.

CO1

Calculate electrical power, energy consumption, tariff charges and apply energy-conservation practices.

Explain conversion of electrical energy into light, heat, chemical and mechanical forms.

CO3

Analyse simple series, parallel and air-gap magnetic circuits and interpret B-H characteristics.

CO4

Explain BJT operating regions, configurations, current gains and design basic bias networks for a specified operating point.

Module roadmap

Module I
Electrical energy consumption and conservation

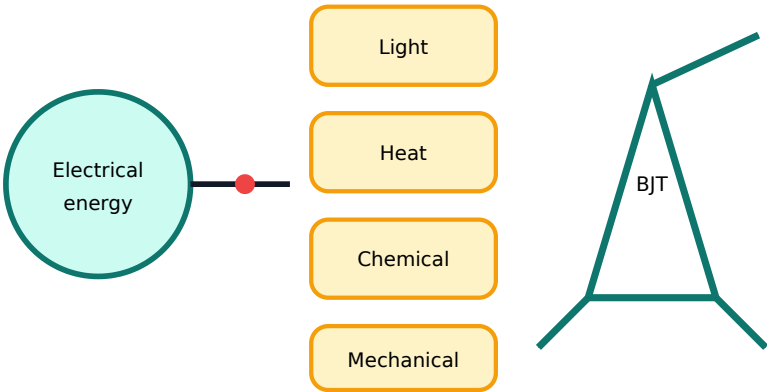
Module II
Applications and conversion of electrical energy

Module III
Magnetism, magnetic circuits and B-H characteristics

Module IV
Bipolar junction transistor and biasing

VISUAL CONCEPT

See the subject as a system



MODULE I

Electrical energy consumption and conservation

Power ഒരു സമയത്ത് ഉപയോഗിക്കുന്ന rate ആണ്; Energy ആ power എത്ര സമയം ഉപയോഗിച്ചു എന്നതിന്റെ ഫലമാണ്. Electricity bill kWh അടിസ്ഥാനത്തിലാണ്; watt മാത്രം നോക്കി monthly charge കണക്കാക്കാൻ കഴിയില്ല.

Power, energy and units

Define power and energy and relate them by $E = Pt$. Use watt, kilowatt, joule, watt-hour and kilowatt-hour correctly.

Daily and monthly consumption

For each appliance multiply rated power by operating hours; add energy of all loads and convert Wh to kWh.

Electricity-bill calculation

Apply tariff slabs, fixed charge, energy charge and other given components exactly as stated. Do not invent current tariff values; use the supplied tariff table.

Importance of conservation

Lower energy consumption reduces cost, peak demand, fuel use and environmental impact without reducing required service.

Domestic conservation practices

Avoid standby loss, choose correct appliance capacity, maintain refrigerator seals, use efficient lighting and operate loads during suitable periods.

BEE star rating

Interpret star level, annual energy consumption and label conditions. Compare appliances using comparable size and service requirements, not stars alone.

Selecting efficient appliances

Consider efficiency, annual energy, lifecycle cost, capacity, usage pattern, maintenance and safety certification.

Rules / common mistakes

- Convert W to kW before multiplying by hours for kWh.
- Use actual operating time, not 24 hours by default.
- Compare appliances of similar capacity.
- Tariff is an input to the problem and may change.

Practice and self-learning

- Prepare a one-day and one-month household load schedule.
- Calculate a sample bill from a supplied tariff table.
- Compare two appliances using BEE label data and lifecycle energy.

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MODULE II

Applications and conversion of electrical energy

15

CO2

Electrical energy light, heat, chemical, mechanical energy ആയി convert ചെയ്യാം. ഓരോ application-ലും conversion principle, losses, efficiency, control, safety എന്നിവ വേർതിരിച്ച് മനസ്സിലാക്കണം.

Lighting conversion

Explain construction and working of incandescent, fluorescent and LED lamps and compare efficacy, life, starting, control and energy use.

Resistance heating

Heat is produced by I^2Rt . Explain ovens, irons and water heaters, heating-element materials and temperature control.

Arc heating

An electric arc produces very high temperature; describe furnaces and welding applications with safety precautions.

Induction heating

Alternating magnetic field induces eddy currents in conductive workpieces. Explain rapid, clean and localised heating applications.

Dielectric heating

High-frequency electric field produces dielectric loss in insulating materials; describe industrial drying and bonding uses.

Electrolysis and Faraday laws

Explain electrolyte, electrodes and ion movement. State Faraday's laws and applications in electroplating, metal refining and battery charging.

Solenoid and relay

Current in a coil creates magnetic force. Explain construction, working, contacts, coil rating and applications.

DC motor

Explain force on a current-carrying conductor, commutation at an elementary level and common applications.

Rules / common mistakes

- Select conversion method based on material and required temperature.

- Heating equipment requires insulation, earthing and temperature protection.

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- Relay coil and contact ratings are different.
- Observe polarity and ventilation in electrochemical work.

Practice and self-learning

- Compare incandescent, fluorescent and LED lamps.
- Prepare a table comparing four electric-heating methods.
- Draw working diagrams of solenoid, relay and elementary DC motor.

MODULE III

Magnetism, magnetic circuits and B-H characteristics

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CO3

Electric circuit-ൽ current flow-നെ resistance oppose ചെയ്യുന്നതുപോലെ magnetic circuit-ൽ flux-നെ reluctance oppose ചെയ്യുന്നു. Air gap ചെറിയതായാലും reluctance വളരെ കൂടാം, കാരണം air permeability കുറവാണ്.

Magnetic quantities

Define magnetic field, flux Φ , flux density B , field strength H , absolute permeability μ and relative permeability μ_r with units.

Magnetic materials

Differentiate diamagnetic, paramagnetic and ferromagnetic materials and identify engineering applications.

MMF and reluctance

Magnetomotive force $F = NI$ drives flux; reluctance $\mathcal{R} = l/(\mu A)$ opposes it. Use $\Phi = F/\mathcal{R}$ as the magnetic-circuit analogue of Ohm's law.

Electric versus magnetic circuits

Compare emf/MMF, current/flux, resistance/reluctance and conductivity/permeability while noting that magnetic circuits are nonlinear.

Series magnetic circuits

For a common flux path, add reluctances and compute required ampere-turns. Handle different materials and cross-sections.

Air-gap circuits

Include gap reluctance using μ_0 ; neglect fringing only when allowed. Air gap usually dominates required MMF.

Parallel magnetic circuits

Polytechnic Study Hub

B-H curve and hysteresis

polypmna.dpdns.org

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Common MMF appears across parallel branches and total flux is the sum of branch fluxes.

Explain initial magnetisation, saturation, retentivity, coercivity and loop area. Relate loop area to hysteresis loss and material selection.

Rules / common mistakes

- Use metres and square metres in reluctance calculations.
- Check whether flux is common or divides.
- Use correct permeability for each section.
- Do not assume constant μ in saturation unless the problem permits it.

Practice and self-learning

- Identify magnetic materials in transformers, motors and relays.
- Solve series and air-gap magnetic circuits.
- Solve one parallel magnetic circuit and interpret a hysteresis loop.

MODULE IV

15

CO4

Bipolar junction transistor and biasing

BJT active region-ൽ amplifier ആയി, cut-off/saturation regions-ൽ switch ആയി ഉപയോഗിക്കാം. Biasing Q-point സ്ഥിരമായി നിലനിർത്താൻ വേണ്ടതാണ്; β മാത്രം ആശ്രയിക്കുന്ന fixed bias stability കുറവായിരിക്കും.

Operating regions

Explain cutoff, active and saturation regions using junction bias conditions and their use in switching/amplification.

Current gains α and β

Define $\alpha = I_C/I_E$ and $\beta = I_C/I_B$. Use $\beta = \alpha/(1-\alpha)$ and $\alpha = \beta/(\beta+1)$, with $I_E = I_C + I_B$.

BJT characteristics

Explain input, output and transfer characteristics and locate cutoff, active, saturation and breakdown regions.

Common-base configuration

Low input resistance, high output resistance, current gain less than one and good high-frequency response.

Common-emitter configuration

Moderate input/output resistance, high current/voltage/power gain and 180° phase

Common-collector configuration

High input resistance, low output resistance, voltage gain near one; emitter follower used for

reversal; widely used amplifier configuration.
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buffering.

Need for biasing and stability

Set a suitable Q-point and prevent large shift with temperature, leakage current and β variation.

Fixed and collector-to-base bias

Compute resistor values for a specified operating point and compare feedback/stability.

Voltage-divider bias

Use divider voltage and emitter resistance to reduce dependence on β ; calculate base, emitter and collector quantities.

Stability factor

Explain the concept and why a lower stability factor is preferred.

Rules / common mistakes

- State the assumed silicon V_{BE} when solving.
- Check that calculated Q-point lies in active region.
- Do not treat β as a precise constant.
- Include emitter resistor voltage when analysing divider bias.

Practice and self-learning

- Plot a qualitative CE characteristic and identify operating regions.
- Compare CB, CE and CC in a table.
- Design fixed-bias, collector-feedback and voltage-divider-bias resistors for a given Q-point.

QUICK REVISION

Formula / rule bank

Rule / Formula 1

$$E = Pt$$

Rule / Formula 2

$$\text{Energy (kWh)} = \text{Power (kW)} \times \text{time (h)}$$

Rule / Formula 3

Heating: $H = I^2 R t$

Rule / Formula 4

Faraday electrolysis: $m \propto It$; $m = ZIt$

Rule / Formula 5

$B = \Phi/A$

Rule / Formula 6

$H = NI/l$

Rule / Formula 7

MMF $F = NI$

Rule / Formula 8

Reluctance $\mathfrak{R} = l/(\mu A)$

Rule / Formula 9

Flux $\Phi = NI/\mathfrak{R}$

Rule / Formula 10

$\beta = IC/IB$; $\alpha = IC/IE$

Rule / Formula 11

$\beta = \alpha/(1-\alpha)$; $\alpha = \beta/(\beta+1)$

Rule / Formula 12

$IE = IC + IB$

Rule / Formula 13

Fixed bias: $R_B \approx (V_{CC} - V_{BE})/I_B$

Rule / Formula 14

Voltage-divider approximation: $V_B = V_{CC} R_2/(R_1 + R_2)$; $V_E = V_B - V_{BE}$

Worked checkpoints

Worked checkpoint 1

A 1.5 kW heater used 2 h/day for 30 days consumes 90 kWh.

Worked checkpoint 2

A magnetic path with $NI = 800$ At and reluctance 2×10^6 At/Wb has flux 0.4 mWb.

Worked checkpoint 3

For $\beta = 99$, $\alpha = 99/100 = 0.99$.

Worked checkpoint 4

For V_{CC} 12 V, desired I_C 2 mA and V_{CE} 6 V with emitter at 1 V, $R_C \approx (12 - 6 - 1)/2 \text{ mA} = 2.5 \text{ k}\Omega$.

APPLICATION

Practical, drawing and self-learning work

No	Task	Required evidence
1	Household load survey and energy schedule.	Record procedure, observation/result and safety/quality checks.
2	Sample tariff/bill calculation.	Record procedure, observation/result and safety/quality checks.
3	BEE-label appliance comparison.	Record procedure, observation/result and safety/quality checks.
4	Lamp comparison and energy-conversion chart.	Record procedure, observation/result and safety/quality checks.
5	Heating-method comparison.	Record procedure, observation/result and safety/quality checks.
6	Electrolysis applications study.	Record procedure, observation/result and safety/quality checks.
7	Solenoid, relay and DC-motor working diagrams.	Record procedure, observation/result and safety/quality checks.
8	Series magnetic-circuit problems.	Record procedure, observation/result and safety/quality checks.
9	Air-gap and parallel magnetic-circuit problems.	Record procedure, observation/result and safety/quality checks.
10	B-H/hysteresis interpretation.	Record procedure, observation/result and safety/quality checks.
11	BJT characteristic-region identification.	Record procedure, observation/result and safety/quality checks.
12	Bias-network numerical design.	Record procedure, observation/result and safety/quality checks.

EXAM PREPARATION

Expected questions and model-answer guides

These are syllabus-aligned practice questions, not a claim about the exact end-semester paper.

2 marks

Q1. Differentiate electrical power and energy with units.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q2. Calculate monthly energy consumption from an appliance schedule.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q3. Explain electricity-bill calculation using a given tariff.

Model-answer guide

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Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q4. List energy-conservation practices and interpret BEE labels.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q5. Compare incandescent, fluorescent and LED lamps.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q6. Compare resistance, arc, induction and dielectric heating.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q7. State Faraday's laws and applications of electrolysis.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q8. Explain solenoid, relay and DC motor principles.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q9. Define magnetic flux, flux density, field strength, permeability, MMF and reluctance.

Model-answer guide

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Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q10. Solve a series magnetic circuit with an air gap.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q11. Explain B-H curve and hysteresis loop.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q12. Explain cutoff, active and saturation regions of a BJT.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q13. Derive the relationship between α and β .

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q14. Compare CB, CE and CC configurations.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q15. Explain the need for biasing and design a voltage-divider bias circuit.

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

COVERAGE AUDIT

Official syllabus coverage checklist

- Module I: Power, energy and units
- Module I: Daily and monthly consumption
- Module I: Electricity-bill calculation
- Module I: Importance of conservation
- Module I: Domestic conservation practices
- Module I: BEE star rating
- Module I: Selecting efficient appliances
- Module II: Lighting conversion
- Module II: Resistance heating
- Module II: Arc heating
- Module II: Induction heating
- Module II: Dielectric heating
- Module II: Electrolysis and Faraday laws
- Module II: Solenoid and relay
- Module II: DC motor
- Module III: Magnetic quantities
- Module III: Magnetic materials
- Module III: MMF and reluctance
- Module III: Electric versus magnetic circuits
- Module III: Series magnetic circuits
- Module III: Air-gap circuits
- Module III: Parallel magnetic circuits
- Module III: B-H curve and hysteresis
- Module IV: Operating regions
- Module IV: Current gains α and β
- Module IV: BJT characteristics
- Module IV: Common-base configuration
- Module IV: Common-emitter configuration
- Module IV: Common-collector configuration
- Module IV: Need for biasing and stability
- Module IV: Fixed and collector-to-base bias
- Module IV: Voltage-divider bias
- Module IV: Stability factor

Learning resources

1. Electrical Technology, Vol. 1 — B. L. Theraja & A. K. Theraja
2. Principles of Electrical Engineering — V. K. Mehta & Rohit Mehta
3. Principles of Electronics — V. K. Mehta & Rohit Mehta
4. Utilization of Electrical Energy and Traction — J. B. Gupta et al.

Primary authority for scope: supplied official SITTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 2031.